

The Challenge of Manufacturing

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Overview

- Numbers vs data
- What is variation?
- Specifications and rectangular loss function
- Inspection
- The definition of quality
- The Taguchi loss function
- The process principle
- Two types of variation
- An example

Is it always the case that
 $2 + 2 = 4$?

When does
 $2 + 2 \neq 4$?

Numbers

Represent the mathematical plane
NOT generated a process
Values are **fixed**

Data

Represent the real world
Generated by a process
Values **vary**

What is variation?

The degree to which the sequential outputs of a process or system differ.

Craft production

Low volume

Manual

Skilled craftsmen

Not interchangeable parts

Mass production

High volume

Automated

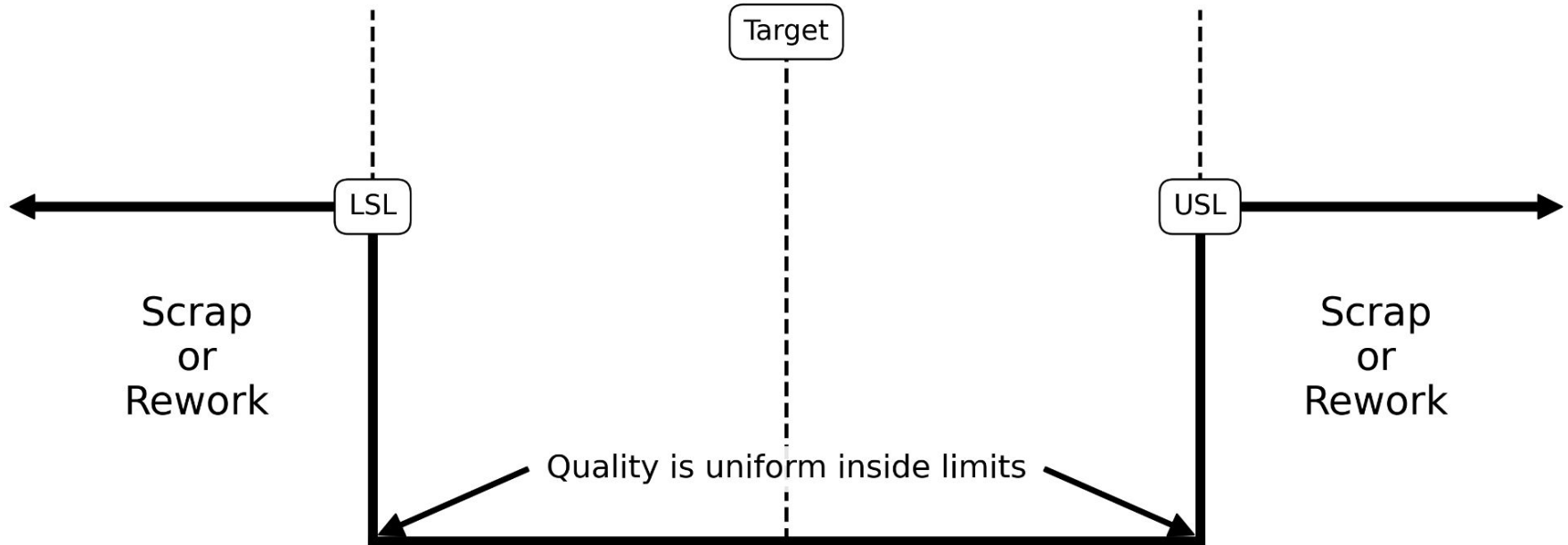
Unskilled labor

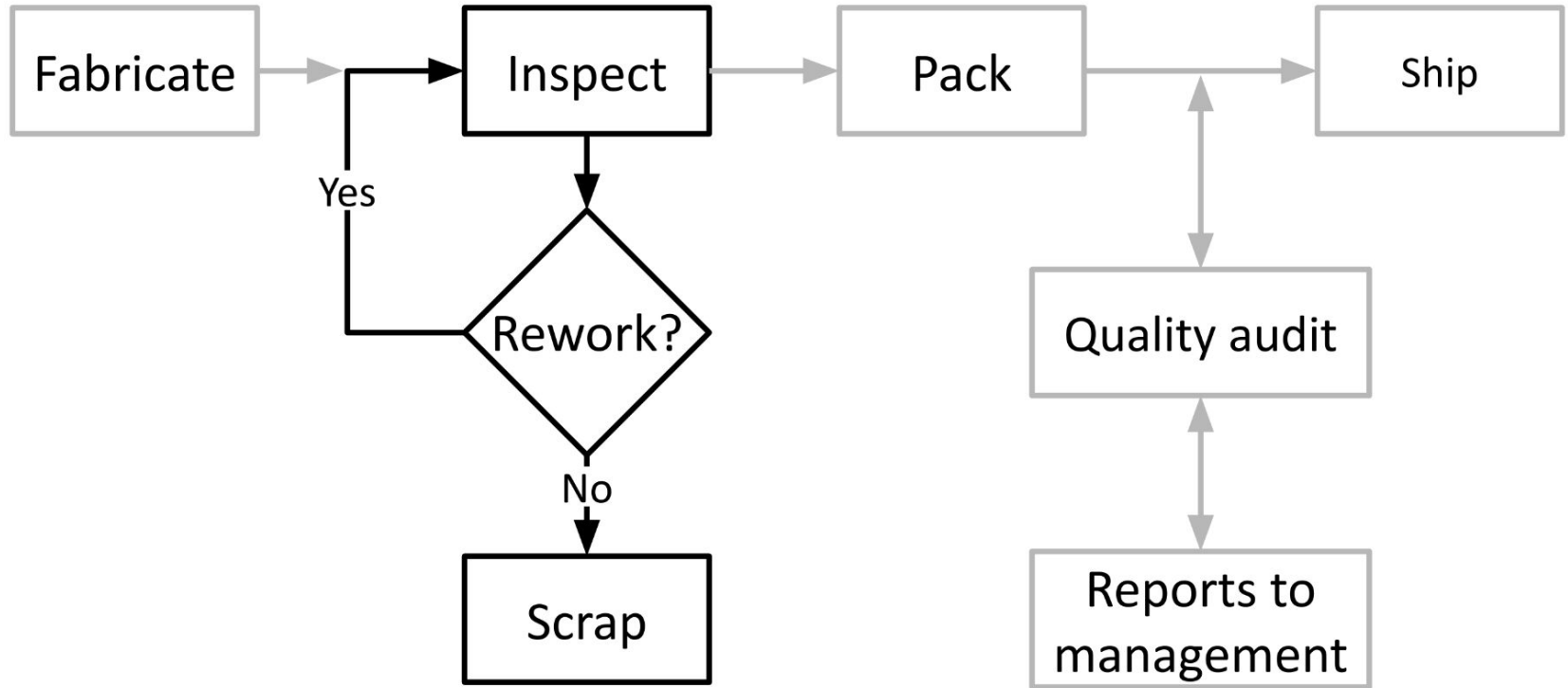
Interchangeable parts

What are specifications?

Specifications define the extent
to which parts can differ
(and still satisfy the customer need).

Rectangular loss function







What is the definition of quality?

On-target with minimum variance.

World-class quality can be approximated with a quadratic loss function of the form:

$$L(x) = K (x - T)^2$$

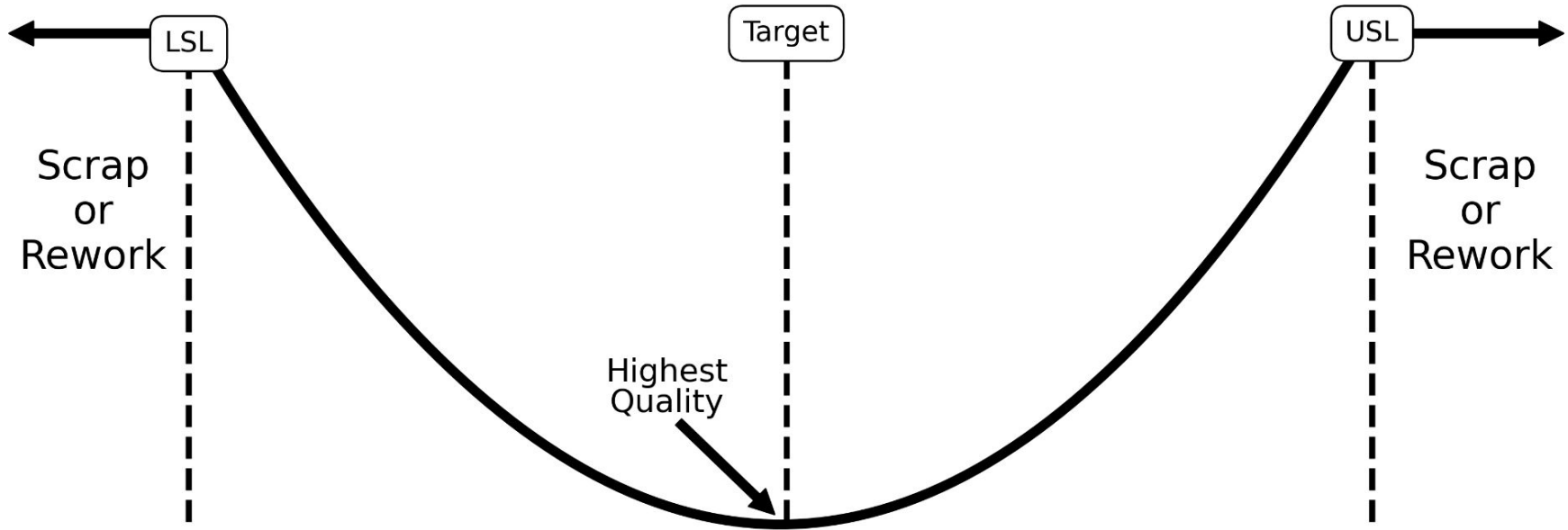
$L(x)$ is the loss function

K is a numeric constant

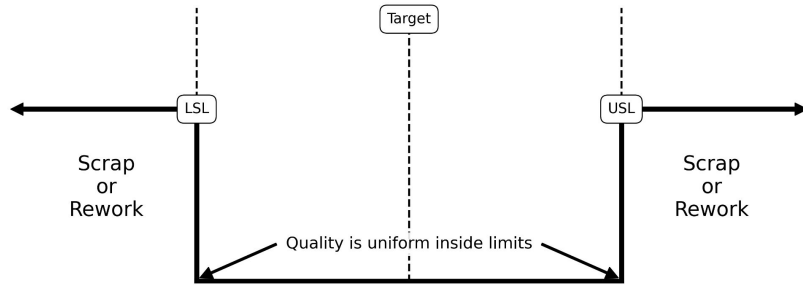
x is the value of the performance or quality characteristic

T is the target value

Taguchi loss function

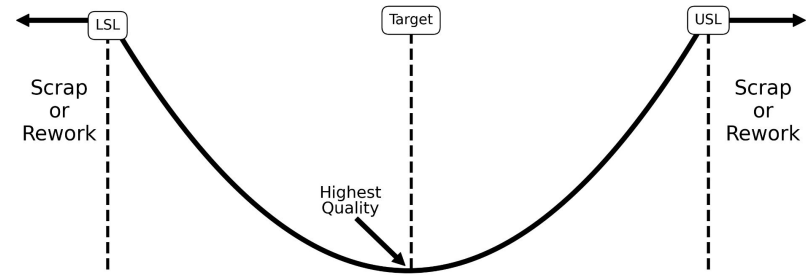


Square loss function



Low fidelity model of loss
Loss occurs **abruptly**
Promotes **reactionary efforts**

Taguchi loss function



High fidelity model of loss
Loss occurs **incrementally**
Promotes **continual improvement**

The process principle:

The sequential outputs of a process are subject to some degree of difference, whether we can measure that difference or not.

Common causes

Produce routine variation

Many in number

Small effect on process behavior

Predictable and **consistent**

Controlled

Assignable causes

Produce exceptional variation

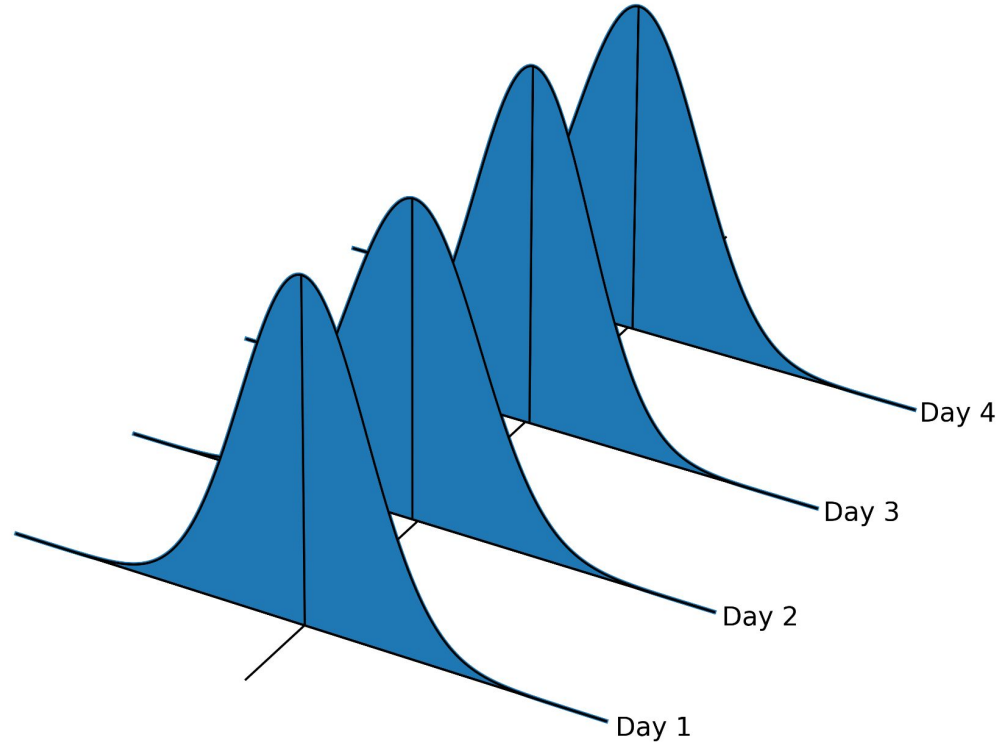
Few in number

Large effect on process behavior

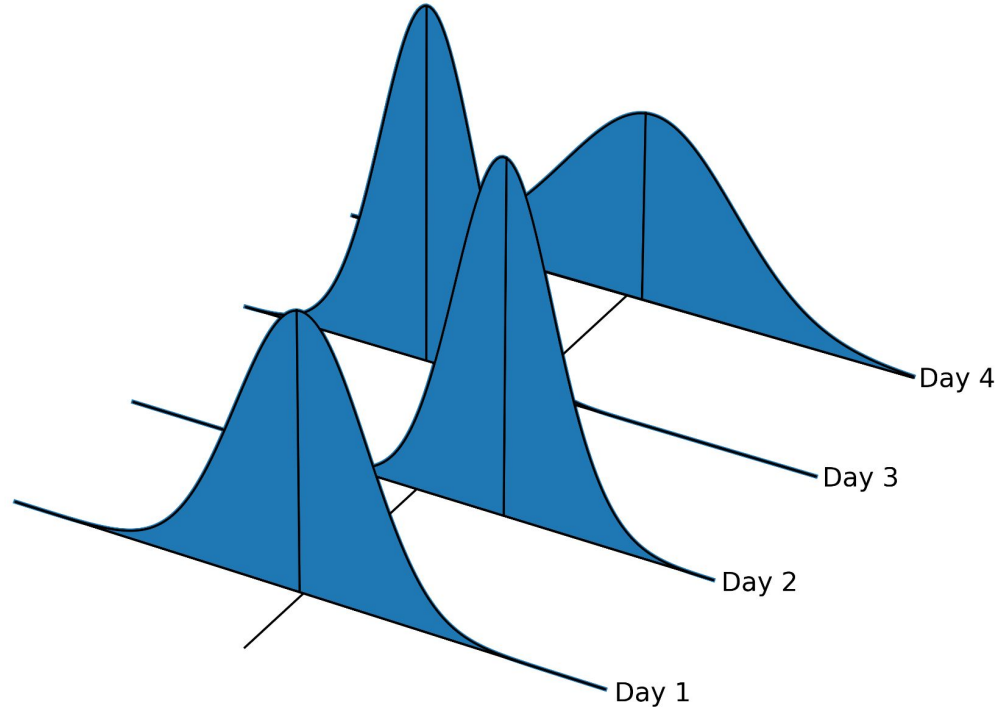
Unpredictable and **inconsistent**

Uncontrolled

What can be said about these distributions?

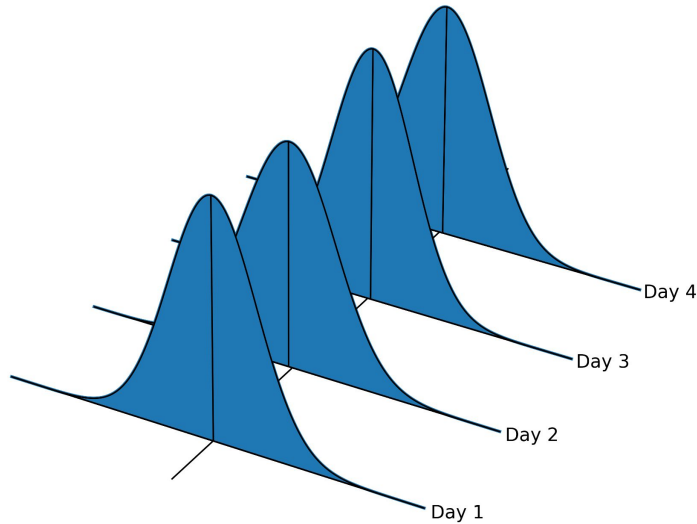


What can be said about these distributions?

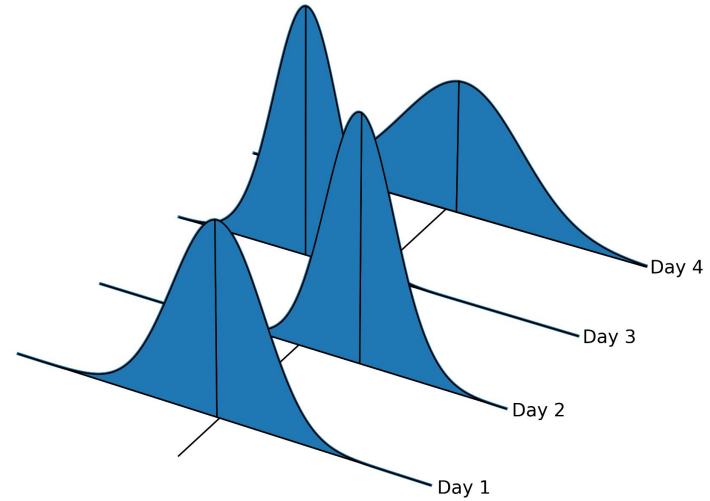


How do we know if a process is predictable or unpredictable?

Predictable process



Unpredictable process



An example

64 electrical resistance measurements

4,400	4,340	4,465	4,260	4,240	4,320	4,310	4,475
4,565	4,500	4,375	4,000	4,350	4,570	4,250	4,725
4,495	4,165	4,660	4,360	4,220	4,380	4,315	4,540
4,325	4,000	4,260	4,500	4,150	4,450	4,380	4,565
4,385	4,125	4,675	4,460	4,520	4,340	4,490	4,210
4,545	4,445	4,800	4,360	4,545	4,350	4,240	4,540
4,235	4,775	4,410	4,265	4,650	4,600	4,350	4,600
4,370	4,615	4,675	4,635	4,230	4,220	4,320	4,850

Measures of location

Mean

Median

Mode

Midrange

Measures of dispersion

```
graph TD; A[Measures of dispersion] --- B[Range]; A --- C[Standard deviation]; A --- D[Interquartile range (IQR)]; A --- E[Root mean square];
```

Range

Standard
deviation

Interquartile
range (IQR)

Root mean
square

Given the summary statistics, what next?

4,400	4,340	4,465	4,260	4,240	4,320	4,310	4,475
4,565	4,500	4,375	4,000	4,350	4,570	4,250	4,725
4,495	4,165	4,660	4,360	4,220	4,380	4,315	4,540
4,325	4,000	4,260	4,500	4,150	4,450	4,380	4,565
4,385	4,125	4,675	4,460	4,520	4,340	4,490	4,210
4,545	4,445	4,800	4,360	4,545	4,350	4,240	4,540
4,235	4,775	4,410	4,265	4,650	4,600	4,350	4,600
4,370	4,615	4,675	4,635	4,230	4,220	4,320	4,850

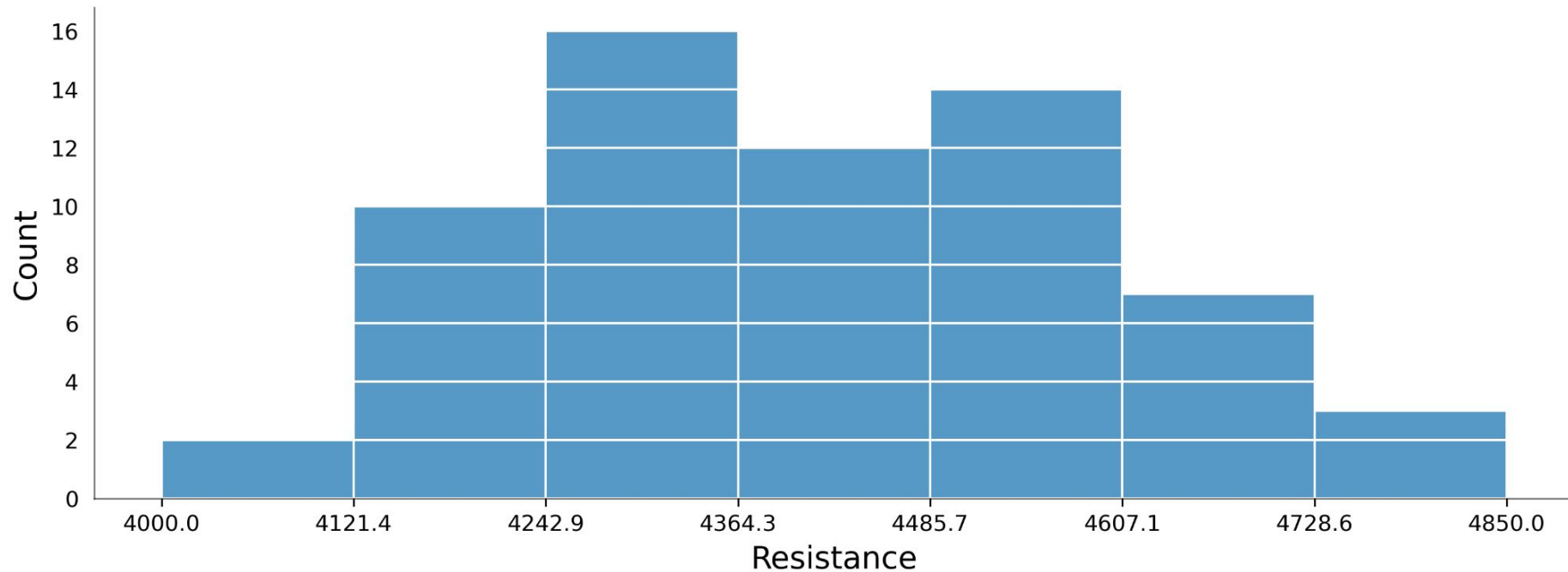
Mean:
4417.8

Median:
4383

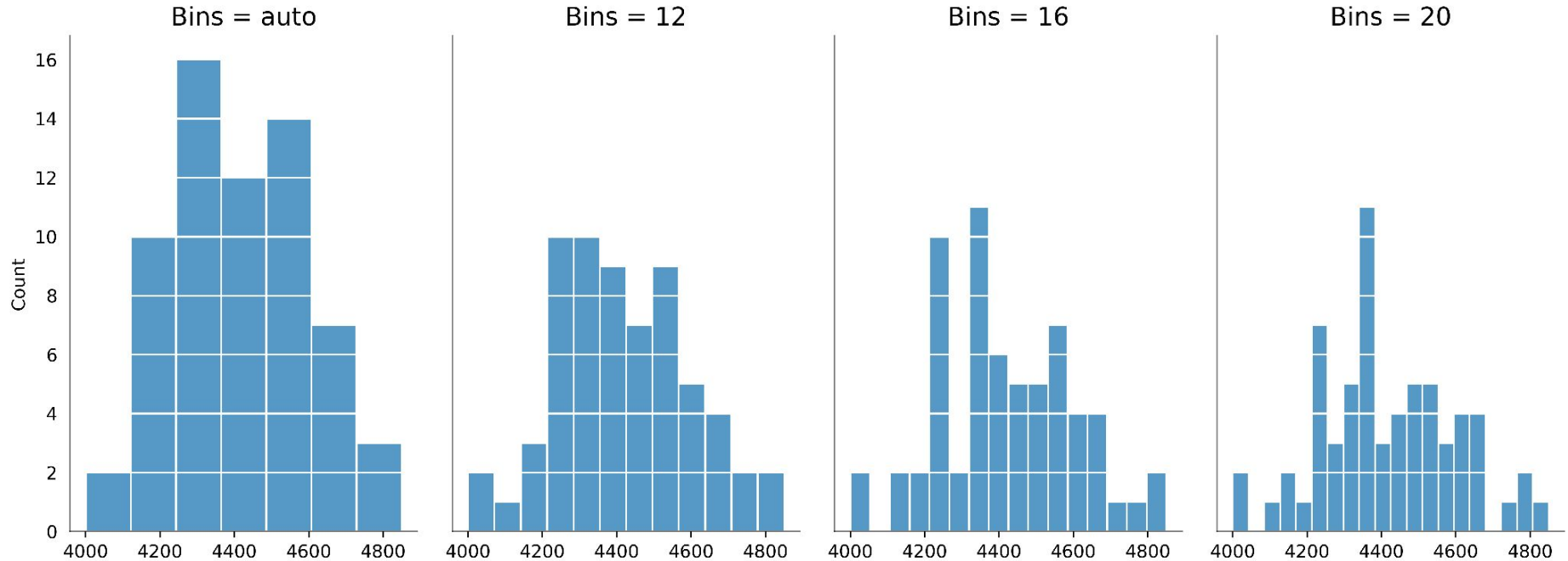
Range:
850

Standard deviation:
185.4

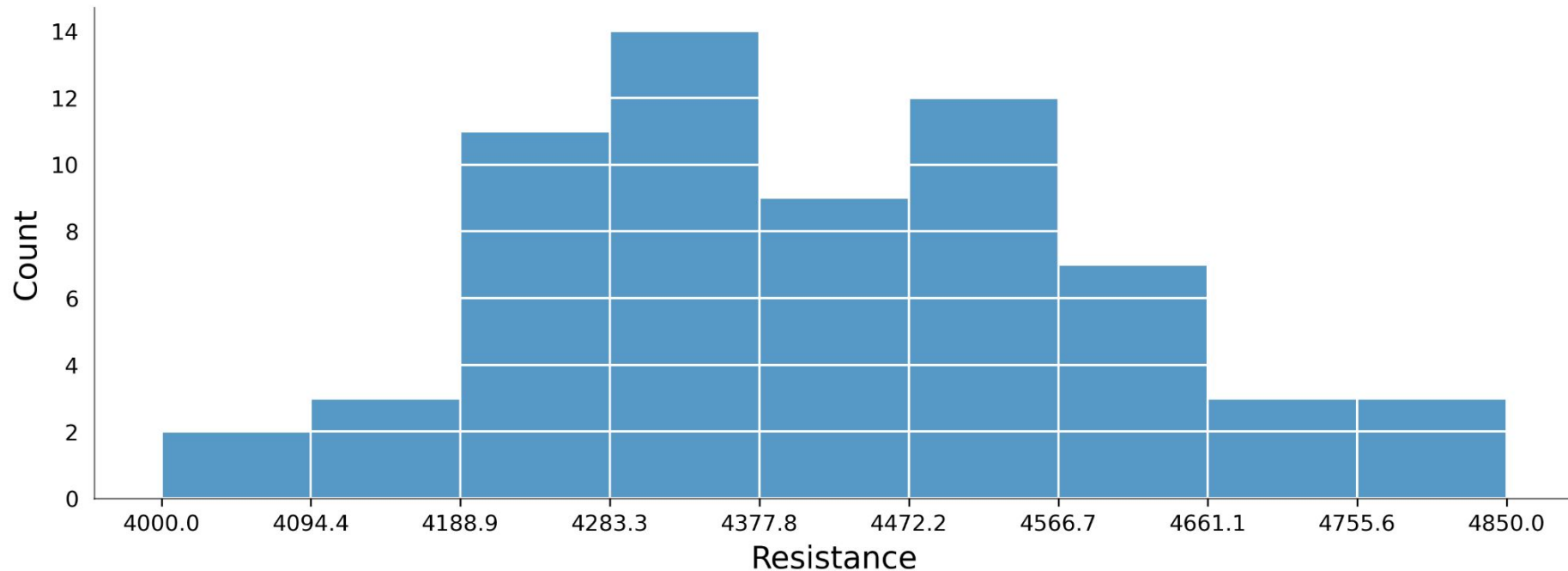
Given the histogram, what next?

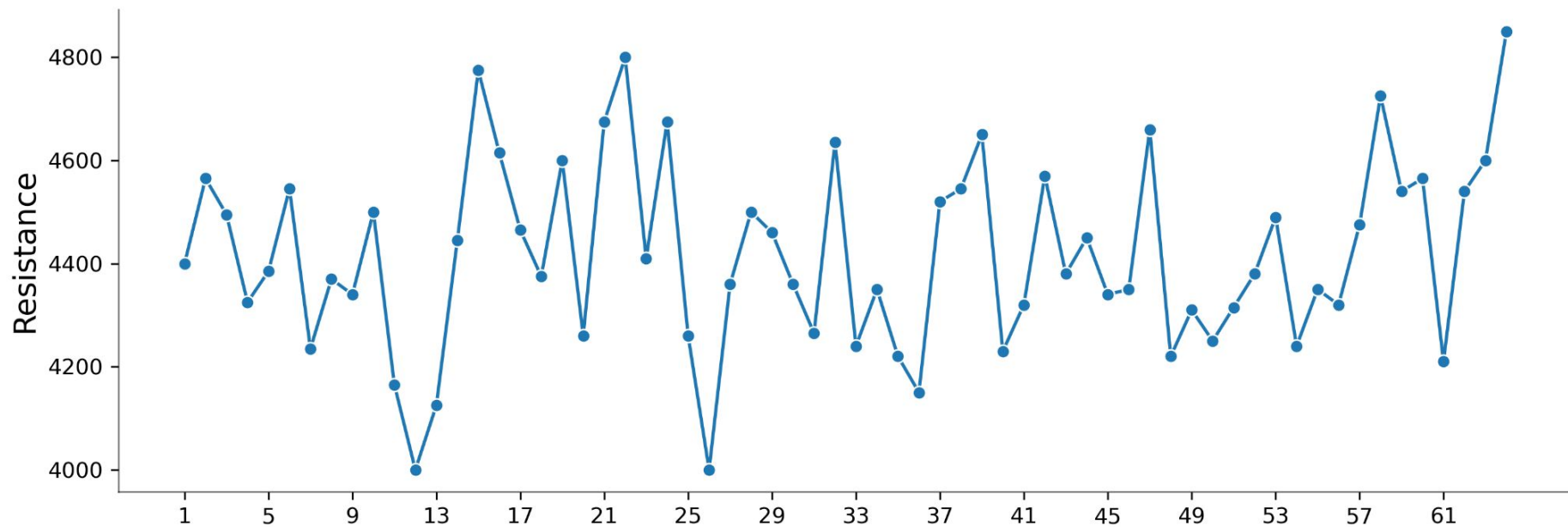


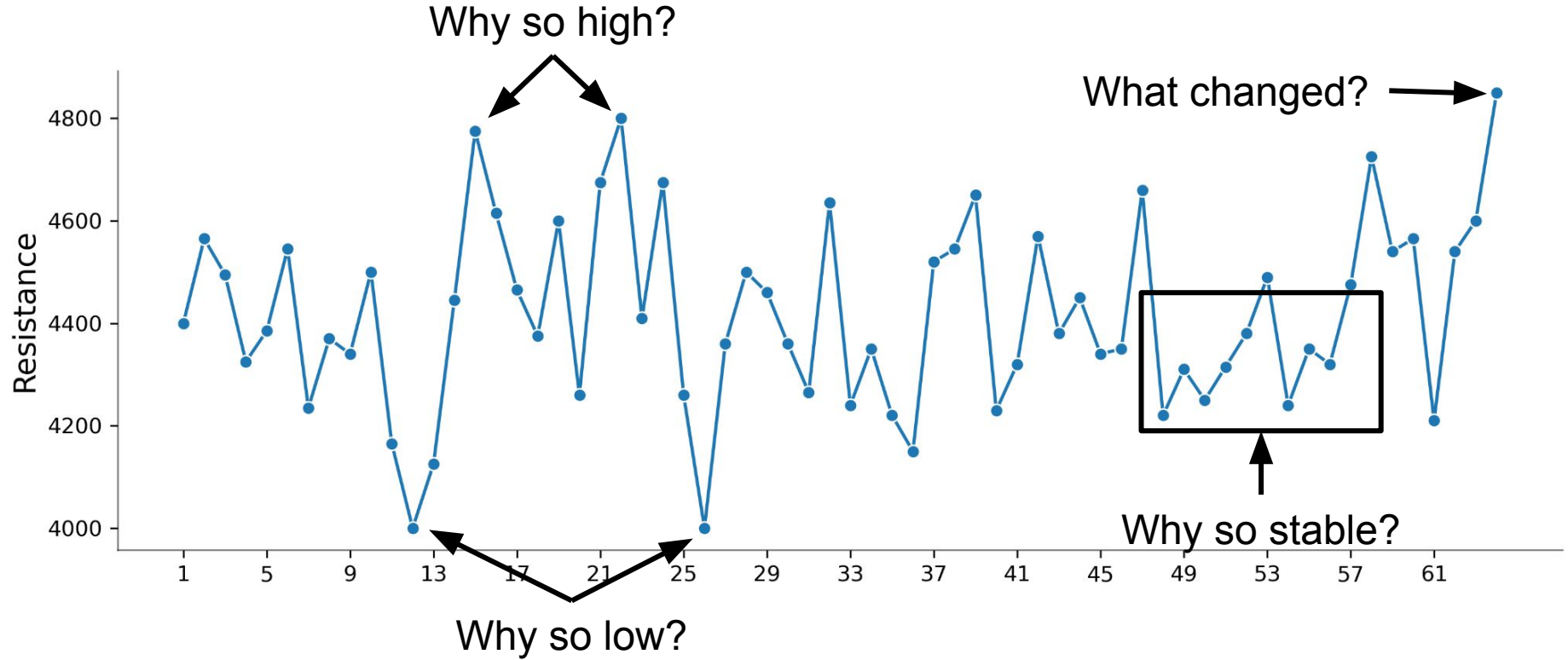
How many bins should we use?



Given the histogram with 9 bins, what next?

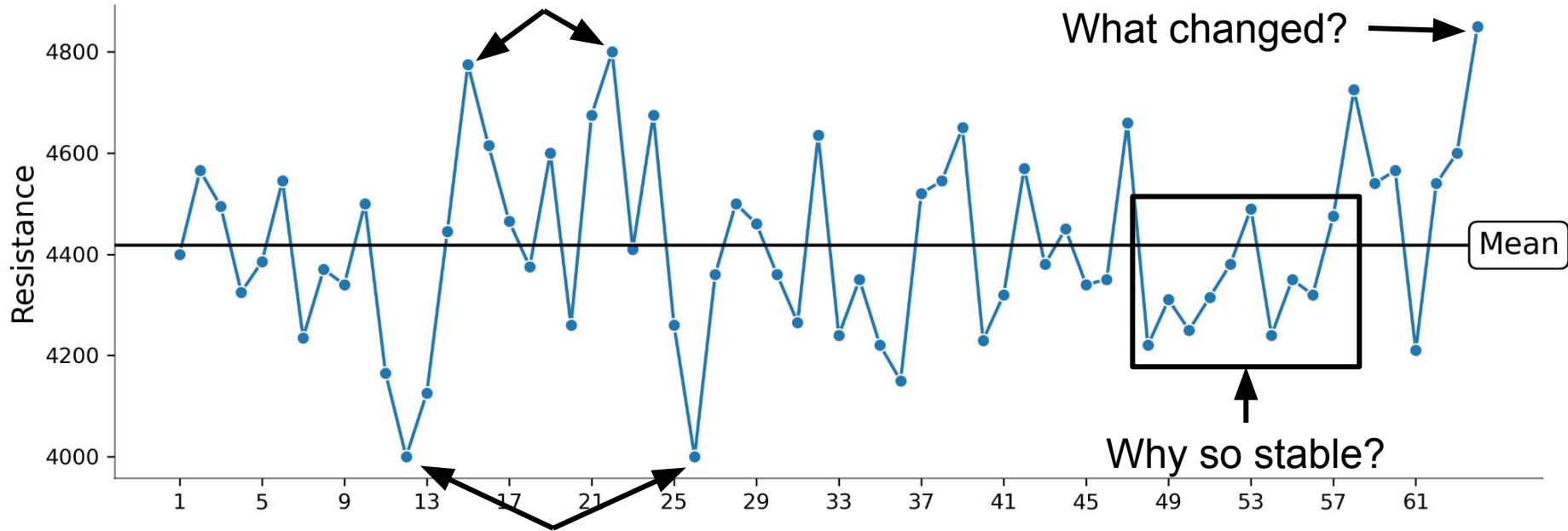






What about including the mean?

Why so much higher than the mean?



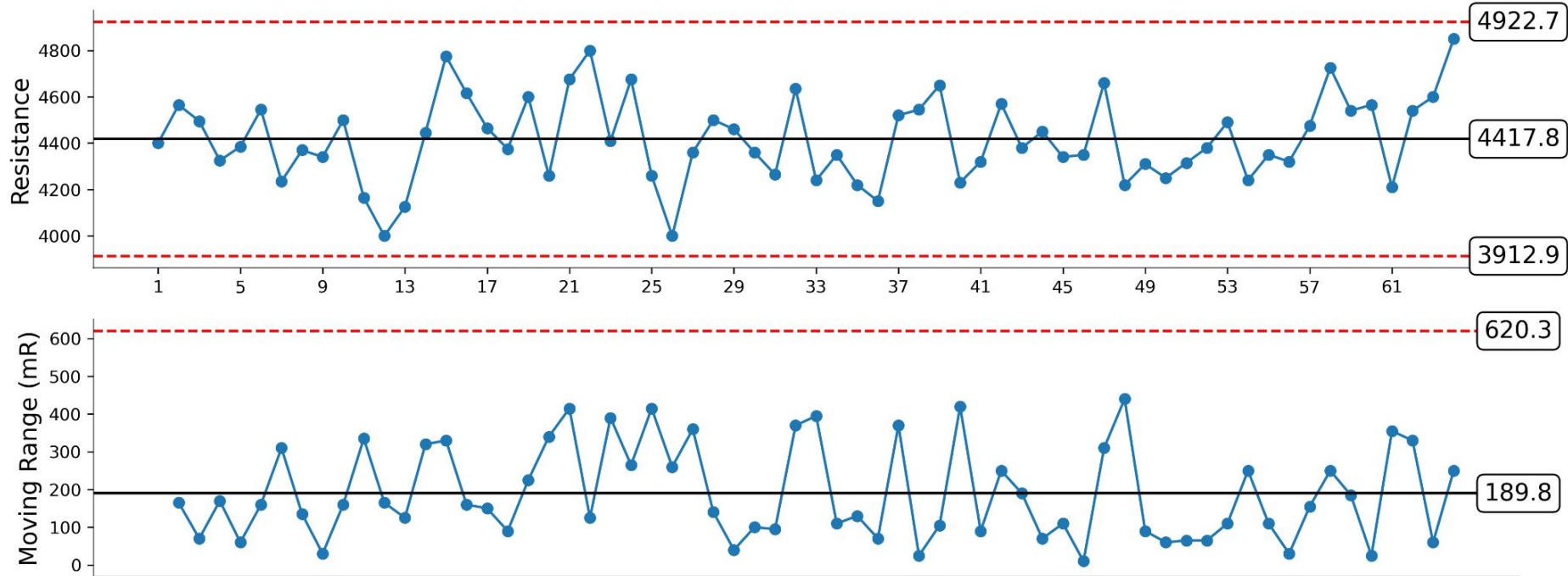
Why so much lower than the mean?

Shewhart's solution:

Average $\pm 3 \cdot \text{Sigma}$

Note: Sigma is NOT the standard deviation statistic. It is one of several within subgroup measures of dispersion that change with subgroup size.

XmR chart of 64 resistance measurements



Process limits for an XmR chart:

$$\text{UPL} = \text{Average} + (2.660 \cdot \text{Ave. mR})$$

$$\text{LPL} = \text{Average} - (2.660 \cdot \text{Ave. mR})$$

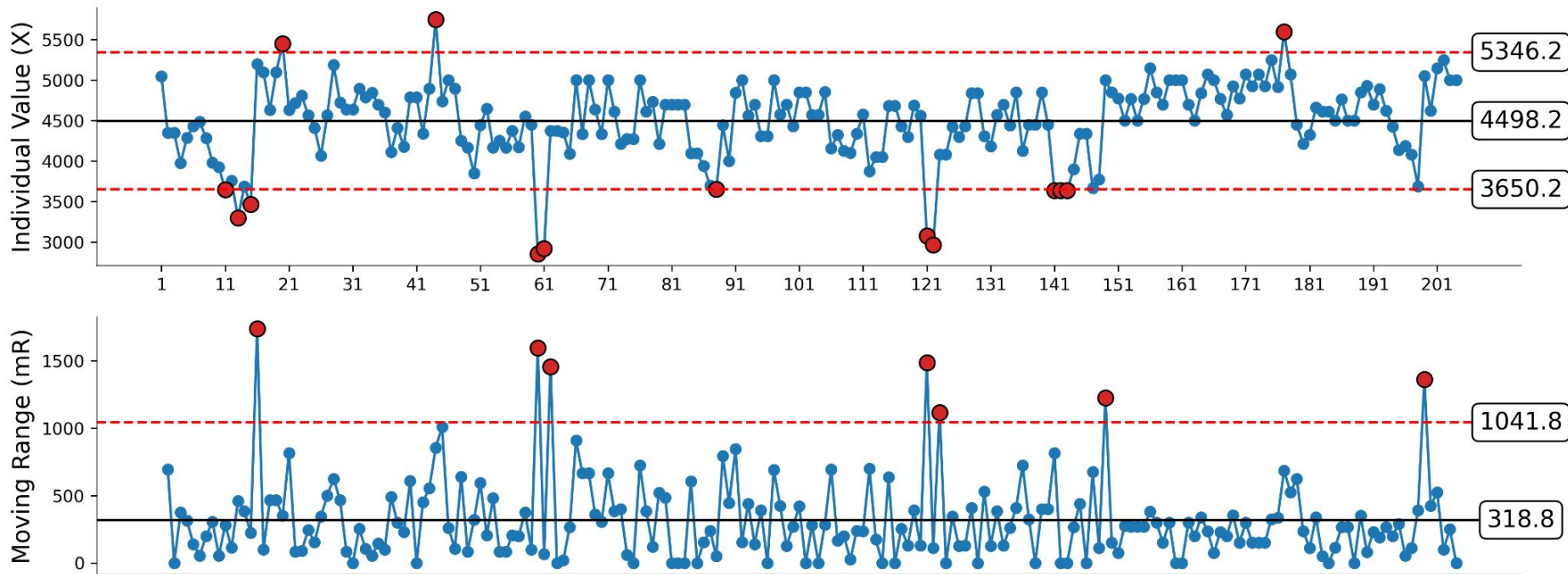
$$\text{URL} = 3.268 \cdot \text{Ave. mR}$$

204 electrical resistance measurements

5,045	4,635	4,700	4,650	4,640	3,940	4,570	4,560	4,450	4,500	5,075	4,500
4,350	5,100	4,600	4,170	4,335	3,700	4,570	3,075	4,450	4,770	4,925	4,850
4,350	5,450	4,110	4,255	5,000	3,650	4,855	2,965	4,850	5,150	5,075	4,930
3,975	4,635	4,410	4,170	4,615	4,445	4,160	4,080	4,450	4,850	4,925	4,700
4,290	4,720	4,180	4,375	4,215	4,000	4,325	4,080	3,635	4,700	5,250	4,890
4,430	4,810	4,790	4,175	4,275	4,845	4,125	4,425	3,635	5,000	4,915	4,625
4,485	4,565	4,790	4,550	4,275	5,000	4,100	4,300	3,635	5,000	5,600	4,425
4,285	4,410	4,340	4,450	5,000	4,560	4,340	4,430	3,900	5,000	5,075	4,135
3,980	4,065	4,895	2,855	4,615	4,700	4,575	4,840	4,340	4,700	4,450	4,190
3,925	4,565	5,750	2,920	4,735	4,310	3,875	4,840	4,340	4,500	4,215	4,080
3,645	5,190	4,740	4,375	4,215	4,310	4,050	4,310	3,665	4,840	4,325	3,690
3,760	4,725	5,000	4,375	4,700	5,000	4,050	4,185	3,775	5,075	4,665	5,050
3,300	4,640	4,895	4,355	4,700	4,575	4,685	4,570	5,000	5,000	4,615	4,625
3,685	4,640	4,255	4,090	4,700	4,700	4,685	4,700	4,850	4,770	4,615	5,150
3,463	4,895	4,170	5,000	4,700	4,430	4,430	4,440	4,775	4,570	4,500	5,250
5,200	4,790	3,850	4,335	4,095	4,850	4,300	4,850	4,500	4,925	4,765	5,000
5,100	4,845	4,445	5,000	4,095	4,850	4,690	4,125	4,770	4,775	4,500	5,000



XmR chart of 204 resistance measurements



Characterization:

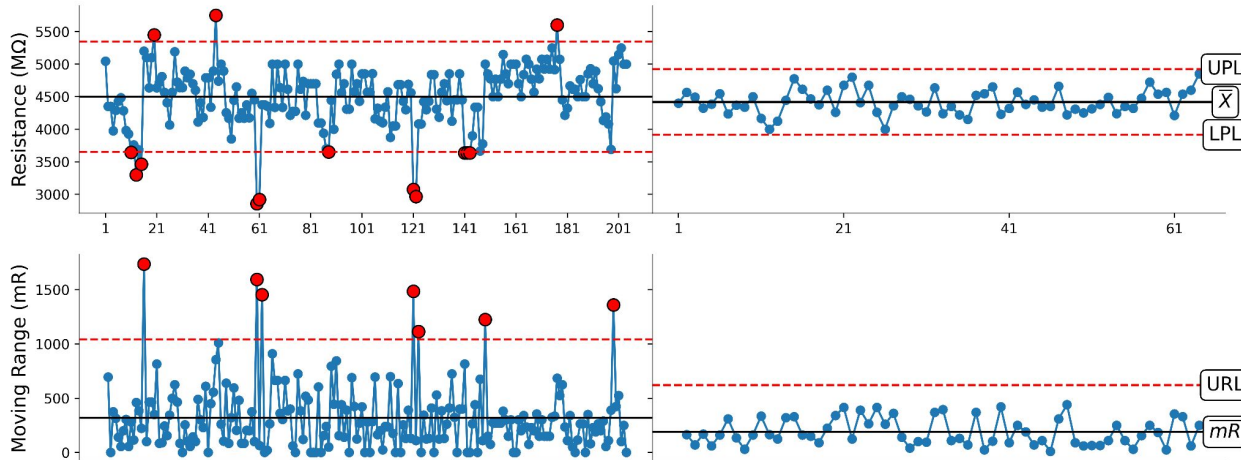
Is all the data
within the process limits?

No

Yes

Process is characterized as
unpredictable

Process is characterized as
predictable



The challenge of manufacturing is...

the challenge of reliably and repeatedly producing products that are, for all intents and purposes, the same.

To reliably and repeatedly produce products that
are the same...

we must know what kinds of variation influence
process behavior and respond in kind.

The only tool capable of this task is the
process behavior chart.

Additional resources

Taguchi loss calculator:

taguchi-loss.streamlit.app

XmR chart web app:

xmr-chart-it.streamlit.app

Questions